



PRESENTED TO THE OFFICE OF MAYOR JOHN CARNEY — CITY OF WILMINGTON, DELAWARE

Mayor John Carney · Office of Economic Development · April 2026

Wilmington's Manufacturing Campus for Sustainable Construction Innovation

A proposal to establish Delaware's first regional CLT/SIP construction manufacturing hub and workforce training incubator — anchored by two active Wilmington projects, creating living-wage jobs, lowering the cost of affordable housing, and positioning Wilmington as the Mid-Atlantic center of the sustainable building economy.

Merge4Ward LLC · Merge 4 Design LLC (M4D) · M4Di™ — Merge 4 Development Institute

· Aubyn Architecture LLC · L.E.E.P., Inc. · Wilmington Housing Authority

Two live Wilmington projects are moving now: Negotiating authorization for a 30+ unit

- ground-up affordable development with a major city agency · Claymont Hybrid CLT/SIP panels arriving May 27 · Building complete mid-July 2026

36

Affordable homes targeted — active ground-up infill negotiation with major city agency. Exceeds client's 30-unit program target by 20%

\$9.2M

Total ROM — active 30+ unit Wilmington infill project, off-site modular. City-agency development budget requiring Catalyst Fund + DSHA coordination

-\$41K

Per-unit subsidy gap (modular). 70% covered by Delaware Catalyst Fund. Solvable with city coordination

\$2M

Wells Fargo / Enterprise grant — pending. Mayor's letter of support by May 1 materially strengthens application

80+

Year-3 manufacturing jobs across dual production lines (CLT/SIP + AAC) at \$45K–\$65K+. Vocational track — no 4-year degree required

One Initiative. Two Problems Solved. Both Are Yours to Own.

Wilmington has two well-documented challenges that have resisted piecemeal solutions: a severe shortage of quality affordable housing production — particularly on cleared infill sites that city agencies already control — and a manufacturing employment base that has declined for two decades without a credible replacement sector in sight. The Merge4Ward Manufacturing Campus addresses both, simultaneously, through a single capital-efficient initiative that requires no city appropriation and is ready to move now.

Two Wilmington projects are in motion today that make this proposal concrete rather than aspirational. The Claymont Hybrid Townhouses — panels arriving May 27, complete mid-July — is the built proof of concept for the construction system. And a representative 30+ unit Wilmington infill project illustrates at city-agency scale exactly what the campus makes financially and technically possible for affordable housing production in Wilmington.

Wilmington Wins

- 30+ affordable homes — illustrative Wilmington infill scenario showing what the campus makes possible: 18 ft wide, 1,260 SF homes where existing corridor stock averages 14–16 ft
- 80+ living-wage manufacturing jobs across two production lines by Year 3 — CNC machinists, autoclave operators, structural assembly techs, QC inspectors — all at \$45K–\$65K+, all accessible without a four-year degree, created in a sector Delaware has been losing for twenty years
- Regional manufacturing — both CLT/SIP structural panels and AAC fire-separation panels — eliminates supply chains to Illinois and Florida that currently add \$8–15/SF in freight and 16–24 weeks in lead time to every Wilmington affordable housing project
- A publicly accessible demonstration center and workforce training facility — a civic asset that attracts conferences, press, and investor attention with Wilmington's name attached
- Opportunity Zone tax base activation on an I-95 corridor site currently producing no tax revenue

Delaware Wins

- First state in the Mid-Atlantic with regional supply chains for **both** CLT/SIP structural panels and AAC fire-separation panels — dual competitive advantage for every developer in the DSHA pipeline. Nearest CLT plant: Illinois. Nearest AAC panel plant: Florida. Both gaps close with this campus.
- NMTC-eligible manufacturing investment attracting \$15–40M in private capital to a qualifying New Castle County census tract across two production lines — no state appropriation required
- IBC 2021 mass timber adoption readiness + AAC noncombustible construction alignment — Delaware positioned as a national early mover on both the structural innovation and fire-safety fronts simultaneously
- DEDO Strategic Fund leverage: \$250K–\$1M in state grants matched against federal and philanthropic capital at 10:1 or better — two qualifying manufacturing lines strengthen the job-creation case
- A dual-track workforce pipeline with Delaware Technical and Community College — CNC/timber production AND autoclave/concrete production — creating the broadest possible employer relationships in advanced sustainable manufacturing

SECTION 2 — WHAT THE CAMPUS MAKES POSSIBLE

An Illustrative Example: 30+ Unit Affordable Infill in Wilmington

To make the campus's impact concrete, consider a representative Wilmington infill scenario: a cleared 1.39-acre former school site in an established corridor, zoned R-3, owned by a city agency, targeted for 30+ units of affordable for-sale housing. M4D and M4Di have produced a full predevelopment cost analysis on exactly this type of project. The numbers reveal precisely why the Merge4Ward campus matters — and what changes when its production lines are operational.

CONSTRUCTION SYSTEM	UNITS	\$/UNIT	TOTAL ROM	GAP/UNIT	VERDICT
Stick-Build Conventional wood frame	36	\$409K	\$14.7M	~\$194K	Status quo
Off-Site Modular Factory volumetric — today	36	\$255K	\$9.2M	~\$41K	Better
CLT/SIP + AAC + Modular Wet Wall CAMPUS SCENARIO — WITH MERGE4WARD	36+	~\$265K*	~\$9.5M*	~\$50K*	The Goal
CLT/SIP — Today Without local supply chain	36	\$413K	\$14.9M	~\$198K	Not viable

*Campus scenario estimate: CLT/SIP structural system + AAC party walls + modular wet wall MEP panels. CLT freight premium of \$12/SF eliminated by regional production. AAC panels sourced locally vs. Florida. Net cost competitive with modular — with a far superior, high-performance building system. Gap closeable via Delaware Catalyst Fund at \$35K/unit.

WHAT THE CAMPUS PRODUCES
— FOR THIS PROJECT TYPE

Three Product Lines. One Complete System.

CLT Structural Panels (Line 1)

Floor decks, roof panels — manufactured on I-95 corridor, delivered in 2–4 weeks instead of 16–24 from Illinois. Eliminates \$8–15/SF freight premium.

AAC Party Wall + Floor Panels (Line 2)

4-hour fire-rated, single-wythe separation walls — manufactured in Delaware instead of shipped from Florida. No stud cavity, no rated assembly complexity, noncombustible by nature.

Modular Wet Wall Panels (Campus Assembly)

Factory-assembled plumbing walls — PEX supply, DWV stack, pressure-tested — delivered to site on Day 1 of framing. Pulls MEP off the critical path by 3–4 weeks across all 36 units.

Workforce trained at the campus

installs all three systems — creating a closed loop between manufacturing jobs and construction employment in Wilmington.

THE POINT: Without the Merge4Ward campus, CLT/SIP hybrid construction costs \$413K/unit in this scenario — unviable for affordable housing. With regional manufacturing, the same high-performance system approaches cost parity with conventional modular, while delivering a noncombustible, energy-efficient, regionally manufactured building that stick-frame and out-of-state modular cannot match. The campus does not just create jobs. It changes what Wilmington can build — and at what price.

SECTION 3 — BUILT PROOF OF CONCEPT

Claymont Hybrid – The System Is Already Working

CLAYMONT HYBRID TOWNHOUSES — LIVE TIMELINE

1316–1324 N. Claymont St., Wilmington, DE

4 units · 3-story · CLT/SIP hybrid construction
Client: L.E.E.P., Inc. / Gwenevere Motley
Architecture: M4D + Aubyn Architecture LLC

CLT panel delivery: May 27, 2026

Building complete: Mid-July 2026

Selling price per unit: \$200,000–\$230,000

First CLT/SIP project completed in Wilmington area

WHAT CLAYMONT PROVES

Four Validated Facts

1. Permittable in Wilmington. ZBA variance for CLT/SIP secured by M4D principals Trevor Knight and Leo Lynch — establishing the first regulatory precedent for this building type in New Castle County.

2. Cost-predictable. M4Di's Gate O ROM came in at \$330/SF locked. Construction actuals: \$358.50/SF. Variance: +5.0% — within the ±15% target range. The methodology works.

3. Affordable. \$200K–\$230K/unit selling price — homeownership in Wilmington at a price that competes with the rental market.

4. Replicable. The IBC Type III–A compliance pathway, the ZBA precedent, and the documented unit cost are all transferable to the active infill engagement and every project after it.

Merge4Ward Manufacturing Campus – Two Production Lines, Four Functions, One Site

The campus requires approximately 60,000–80,000 SF of industrial space on Delaware’s I-95 corridor in a designated Opportunity Zone. What was originally conceived as a single CLT/SIP manufacturing line has evolved — based on active project analysis — into a **dual-line sustainable construction campus**: one line producing CLT structural panels and SIP envelopes, a second producing autoclaved aerated concrete (AAC) party wall and floor panels. Together they manufacture the complete high-performance townhouse system. No facility like this exists anywhere in the Mid-Atlantic. The nearest CLT plant is in Illinois; the nearest AAC panel plant is in Florida. Delaware becomes the regional source for both.

LINE 01 · STRUCTURE CLT + SIP Manufacturing CNC-controlled cross-laminated timber panel fabrication and structurally insulated panel assembly. Eliminates the 2,000-mile Illinois/West Coast supply chain. Reduces per-project freight by \$8–15/SF and lead times from 16–24 weeks to 2–4 weeks. Serves mid-rise affordable housing, community facilities, and adaptive reuse projects across the I-95 corridor.	LINE 02 · SEPARATION AAC Wall + Floor Panels Autoclaved aerated concrete (AAC) party wall and floor panels — the only fire-separation system that delivers a 4-hour UL rating in a single 4" wythe with no stud cavity, no rated assembly, and no combustible framing. Currently manufactured only in Florida (AERCON) with no Mid-Atlantic plant in operation. A Delaware facility is the first in the region, serving every townhouse and mid-rise developer within 500 miles.	FUNCTION 03 · PEOPLE Workforce Training Incubator Integrated training program with Delaware Technical and Community College across both production lines: CNC machinists, autoclave operators, structural assembly technicians, QC inspectors, and construction logistics coordinators. Year-1 cohort: 20 trainees across both lines. Year-3 target: 80+ graduates annually. All positions \$45,000–\$65,000+ — no four-year degree required.	FUNCTION 04 · PLATFORM Innovation Hub & Demo Center A publicly accessible demonstration center showcasing the complete dual-system building technology — CLT/SIP structure alongside AAC fire separation — for architects, developers, housing authorities, and municipal agencies. Positions Wilmington as the Mid-Atlantic center for sustainable construction innovation. Every site visit carries Mayor Carney's name as the civic leader who made it happen.
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WHY TWO LINES ON ONE CAMPUS: CLT/SIP and AAC are complementary manufacturing processes — both require large-format precision cutting (CNC router/saw), controlled-environment production, and crane-assisted panel handling. The shared infrastructure (facility, crane, logistics dock, CNC equipment, workforce) makes a dual-line campus more capital-efficient than two separate facilities. The combined product offering — structural system plus fire-separation system — also makes the campus the single-source supplier for the complete high-performance townhouse package, creating stronger customer relationships and higher order volume per project.

SITE CRITERIA FOR MAYOR'S OFFICE COORDINATION: The expanded dual-line facility requires ~60,000–80,000 SF of industrial space with loading dock access, crane rail clearance, and autoclave infrastructure capacity (AAC curing requires high-pressure steam). Delaware's I-95 corridor has multiple Opportunity Zone–designated industrial parcels meeting these criteria, including former industrial sites with existing utility capacity. DEDO coordinates OZ mapping and can identify city-owned or city-adjacent inventory. The Mayor's Office DEDO introduction is the fastest path to site identification.

SECTION 5 – FUNDING & INCENTIVE ROADMAP

No City Dollars Required – But City Access Unlocks Everything

The campus assembles from a layered capital stack in which each component de-risks the next. The Mayor's Office does not fund the facility — it opens the doors that make the funding stack possible. A single letter of support for the Wells Fargo grant by May 1 sets the entire stack in motion.

FUNDING SOURCE	TYPE	EST. RANGE	CITY LEVERAGE POINT
Enterprise / Wells Fargo Housing Affordability Breakthrough Challenge ★ URGENT	Federal Grant	\$2,000,000	Mayor's letter of support by May 1 materially strengthens the application. Deadline May 15, 2026. This is the catalyst grant that funds the market study and Year-1 workforce cohort. Single most time-sensitive item in this brief.
New Markets Tax Credits (NMTC)	Tax Credit Equity	\$15M-\$40M	Primary facility capitalization. Qualifying Delaware OZ census tract + documented job creation = strong NMTC eligibility. City endorsement and site designation coordination with DEDO accelerates the CDE application.
DEDO Strategic Fund	State Grant	\$250K-\$1M	Manufacturing jobs at \$45K+/year are the primary trigger. Mayor's endorsement and direct DEDO introduction accelerates review. Rolling — apply Q3 2026.
Delaware Catalyst Fund	State Program	\$35K/unit	Directly closes 70% of the Lobdel subsidy gap (\$35K × 36 units = \$1.26M). Lobdel as a WHA/city project is a natural Catalyst Fund candidate. City coordination with DSHA accelerates eligibility determination.
EDA Build to Scale / Regional Technology	Federal Grant	\$500K-\$2M	Advanced manufacturing in economically distressed region. New Castle County census tracts qualify. City letter of support and OZ designation strengthen the application. Annual NOFA — apply Q3 2026.
HUD Economic Development Initiative	Federal Grant	\$250K-\$1M	LMI community economic development + manufacturing job creation. City of Wilmington as co-applicant or support partner directly amplifies the community benefit documentation required for eligibility.
Opportunity Zone + New Castle County TIF	Tax Incentive	Site-specific	Capital gains deferral for private investors. TIF for infrastructure improvements on city-adjacent industrial sites. Both mechanisms require the city's OZ coordination and DEDO alignment to activate.
Longwood Foundation / JPMorgan Chase AdvancingCities	Philanthropic	\$250K-\$1M	Sustainable development, workforce development, economic mobility in underserved communities. Delaware and Wilmington focus aligns strongly. City endorsement strengthens applications.
DOE Building Technologies / Advanced Mfg.	Federal Grant	\$1M-\$5M	CLT manufacturing as advanced building materials production with documented carbon reduction. Requires university

FUNDING SOURCE	TYPE	EST. RANGE	CITY LEVERAGE POINT
			research partner (U of Delaware, Delaware State). Year-2 application with city support.

INDICATIVE TOTAL CAPITAL STACK: \$2M Wells Fargo catalyst + \$15–40M NMTC equity + \$1–4M federal grants (EDA, HUD, DOE) + \$500K–\$2M state grants (DEDO, DEMEP) + \$250K–\$1M philanthropic = **\$19M–\$49M TOTAL CAPITALIZATION** — appropriate for a 60,000–80,000 SF dual-line manufacturing facility (CLT/SIP + AAC). Private equity and debt complete the stack. **THE CITY CONTRIBUTES ZERO DOLLARS AND ENABLES THE ENTIRE RAISE.**

SECTION 6 – DEVELOPMENT ROADMAP

Four Phases – Each Funds the Next

PHASE 1 • 2026 ← NOW Market Study + Site Selection \$500K • Wells Fargo Grant Mid-Atlantic demand analysis for both CLT/SIP and AAC panel markets. Dual-line facility site identification with DEDO on I-95 OZ corridor. DelTech curriculum scoping across both production tracks. NMTC eligibility analysis. Claymont complete — first CLT/SIP and AAC cost data documented in ALDDB.	PHASE 2 • 2027 Facility Build + Workforce Launch \$900K • NMTC + State Grants Dual-line fit-out: CNC router/saw for CLT/SIP Line 1; autoclave and AAC cutting equipment for Line 2. First workforce cohort — 20 trainees across both lines. First regional CLT/SIP and AAC panels delivered to 3 pipeline projects. NMTC closing with CDE partner.	PHASE 3 • 2028 Regional Scale \$600K • Operating Revenue + Grants 8–12 projects/year. Second cohort (40 trainees). Merge4Ward Builder's Manual published — replicable CLT/SIP guide for Mid-Atlantic developers. DOE advanced manufacturing grant application with U of Delaware.	PHASE 4 • 2029+ National Standard-Setting Self-sustaining + licensing Wilmington recognized as Mid-Atlantic center of sustainable construction innovation — CLT/SIP and AAC. Affiliate licensing to Baltimore, Philadelphia, D.C. peers. Academy Level 2 CLT + AAC certification. Phase II and III affordable infill projects underway on additional city-agency sites.
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Four Steps. No City Funds Required. One Deadline.

- **Step 1 — Letter of Support by May 1 (URGENT):** A letter from Mayor Carney's office endorsing the Merge4Ward initiative and supporting the Wells Fargo / Enterprise Housing Affordability Breakthrough Challenge application. Deadline May 15, 2026. This is a \$2M grant that funds the market study and Year-1 workforce training cohort — and the Mayor's letter is the single item that most meaningfully separates a winning application from a strong one. We will provide a ready-to-sign draft.
- **Step 2 — DEDO Introduction:** A warm introduction to DEDO's Strategic Fund program officer and Opportunity Zone coordinator. This accelerates Merge4Ward's site identification and positions the facility for the Q3 2026 DEDO grant application cycle — where the Mayor's prior relationship creates a substantive advantage over cold applications.
- **Step 3 — DSHA / Catalyst Fund Coordination for Lobdel:** A facilitated conversation between the Mayor's Office, WHA, and DSHA to confirm the Delaware Catalyst Fund eligibility pathway for the Lobdel Avenue project. At \$35K/unit × 36 units = \$1.26M, Catalyst Fund closes 70% of the modular scenario gap — making the WHA project financeable without extraordinary subsidy layering.
- **Step 4 — Civic Endorsement as a Wilmington Priority:** Recognition of the Merge4Ward campus as a Wilmington economic development priority in appropriate city communications, press events, and stakeholder settings. This costs nothing. It returns the press coverage, investor attention, and reputational positioning that makes Wilmington — not a neighboring jurisdiction — the address on the first regional CLT manufacturing facility in the Mid-Atlantic.

The Team Behind This Proposal

Merge4Ward LLC Manufacturing entity & initiative lead · Wilmington, DE	Merge 4 Design LLC Architecture · Design · Predevelopment Intelligence Knight · Ainsworth · Lynch · Davis	M4Di™ Predevelopment Intelligence 501(c)(3) · CB CDC fiscal sponsorship	Aubyn Architecture LLC Architecture · AOR David C. Ainsworth AIA · DE/PA/CT	L.E.E.P., Inc. Community Developer · 501(c) (3) Gwenevere Motley · Wilmington, DE	Wilmington Pipeline 30+ unit affordable infill scenario Representative of city-agency project demand
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